PHYDSC403T — Basic Quantum Mechanics

Unit 5: Angular Momentum Operators and Symmetry Transformations

7th Semester, Assam University (FYUG) — Haflong Government College

Paper: PHYDSC403T, Full Marks 100 [ESE 70 + CCA 30] | Unit 5: 13 lectures

Notes 15 of 20 — drafted by gemini-3.6-flash (Kaggle Model Proxy) from the unit syllabus only, without external sources, and checked against the syllabus.

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Unit syllabus (as in the syllabus file)

1. Angular momentum operators, Commutation rules for angular momentum operators.
2. Eigen value and eigen functions of L^2, L_z. Eigen value of total angular momentum.
3. Ladder operator and their different commutation relations.
4. Symmetry transformations: Space time translations and rotations, Invariance under the transformations and conservation laws.

Top 2 high-mark topics in this unit

1. Angular momentum operators, commutation rules, eigenvalues and eigenfunctions of L^2 and L_z, and ladder operators — 13 lectures, joint heaviest unit, derivation-heavy operator algebra, and zero AU coverage so the whole unit is untested risk.
2. Symmetry transformations: space-time translations and rotations, invariance under the transformations and conservation laws — second half of the unit, concept plus derivation, zero AU coverage, and a standard long question pairing with ladder operators.

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7. Questions — Short form, then Long form (no AU questions found for this unit)
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1. Angular Momentum Operators

Definition and Cartesian Components

In classical mechanics, the orbital angular momentum of a particle with position vector $\mathbf{r}$ and linear momentum $\mathbf{p}$ is defined by the vector cross product $\mathbf{L} = \mathbf{r} \times \mathbf{p}$. In quantum mechanics, the corresponding orbital angular momentum operator $\hat{\mathbf{L}}$ is defined by replacing the position vector $\mathbf{r}$ and linear momentum $\mathbf{p}$ with their respective quantum mechanical operators, $\hat{\mathbf{r}}$ and $\hat{\mathbf{p}}$:

$$\hat{\mathbf{L}} = \hat{\mathbf{r}} \times \hat{\mathbf{p}}$$

In Cartesian coordinates, the position operator is written as $\hat{\mathbf{r}} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ and the momentum operator as $\hat{\mathbf{p}} = p_x\mathbf{i} + p_y\mathbf{j} + p_z\mathbf{k}$. Expanding the vector product in determinant form gives:

$$\hat{\mathbf{L}} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ x & y & z \\ p_x & p_y & p_z \end{vmatrix}$$

Evaluating the determinant along the first row gives the Cartesian components of the angular momentum operator:

$$L_x = yp_z - zp_y$$

$$L_y = zp_x - xp_z$$

$$L_z = xp_y - yp_x$$

In the position representation, the Cartesian components of the linear momentum operator are represented by partial spatial derivatives:

$$p_x = -i\hbar\frac{\partial}{\partial x}$$

$$p_y = -i\hbar\frac{\partial}{\partial y}$$

$$p_z = -i\hbar\frac{\partial}{\partial z}$$

Substituting the differential forms of p_y and p_z into the expression for L_x gives:

$$L_x = y\left(-i\hbar\frac{\partial}{\partial z}\right) - z\left(-i\hbar\frac{\partial}{\partial y}\right)$$

Factoring out the constant $-i\hbar$ yields the differential-operator form of L_x:

$$L_x = -i\hbar \left(y\frac{\partial}{\partial z} - z\frac{\partial}{\partial y} \right)$$

Similarly, substituting the differential forms of p_z and p_x into the expression for L_y gives:

$$L_y = z\left(-i\hbar\frac{\partial}{\partial x}\right) - x\left(-i\hbar\frac{\partial}{\partial z}\right)$$

Factoring out $-i\hbar$ yields the differential-operator form of L_y:

$$L_y = -i\hbar \left(z\frac{\partial}{\partial x} - x\frac{\partial}{\partial z} \right)$$

Finally, substituting the differential forms of p_x and p_y into the expression for L_z gives:

$$L_z = x\left(-i\hbar\frac{\partial}{\partial y}\right) - y\left(-i\hbar\frac{\partial}{\partial x}\right)$$

Factoring out $-i\hbar$ yields the differential-operator form of L_z:

$$L_z = -i\hbar \left(x\frac{\partial}{\partial y} - y\frac{\partial}{\partial x} \right)$$

Angular Momentum Operators in Spherical Polar Coordinates

When solving quantum mechanical problems with spherical symmetry, it is convenient to transform the angular momentum operators from Cartesian coordinates (x, y, z) to spherical polar coordinates (r, θ, ϕ). The transformation equations relating Cartesian and spherical polar coordinates are:

$$x = r\sin\theta\cos\phi$$

$$y = r\sin\theta\sin\phi$$

$$z = r\cos\theta$$

To transform the differential operators, we apply the multivariate chain rule for partial derivatives. The partial derivative with respect to the azimuthal angle ϕ is expanded as:

$$\frac{\partial}{\partial\phi} = \frac{\partial x}{\partial\phi}\frac{\partial}{\partial x} + \frac{\partial y}{\partial\phi}\frac{\partial}{\partial y} + \frac{\partial z}{\partial\phi}\frac{\partial}{\partial z}$$

Differentiating the coordinate transformation equations with respect to ϕ gives $\frac{\partial x}{\partial\phi} = -r\sin\theta\sin\phi = -y$, $\frac{\partial y}{\partial\phi} = r\sin\theta\cos\phi = x$, and $\frac{\partial z}{\partial\phi} = 0$. Substituting these partial derivatives into the chain rule expression yields:

$$\frac{\partial}{\partial\phi} = -y\frac{\partial}{\partial x} + x\frac{\partial}{\partial y}$$

Rearranging the terms on the right-hand side gives:

$$\frac{\partial}{\partial \phi} = x\frac{\partial}{\partial y} - y\frac{\partial}{\partial x}$$

Multiplying both sides by $-i\hbar$ gives:

$$-i\hbar\frac{\partial}{\partial \phi} = -i\hbar\left(x\frac{\partial}{\partial y} - y\frac{\partial}{\partial x}\right)$$

Recognizing the right-hand side as the definition of L_z, we obtain the expression for L_z in spherical polar coordinates:

$$L_z = -i\hbar\frac{\partial}{\partial \phi}$$

By applying the chain rule transformations for x, y, z to L_x and L_y, their forms in spherical polar coordinates are obtained as:

$$L_x = i\hbar\left(\sin\phi\frac{\partial}{\partial\theta} + \cot\theta\cos\phi\frac{\partial}{\partial\phi}\right)$$

$$L_y = -i\hbar\left(\cos\phi\frac{\partial}{\partial\theta} - \cot\theta\sin\phi\frac{\partial}{\partial\phi}\right)$$

The magnitude squared of the total orbital angular momentum operator is defined as the scalar product $L^2 = \mathbf{L}\cdot\mathbf{L} = L_x^2 + L_y^2 + L_z^2$. Squaring the spherical coordinate representations of L_x, L_y, and L_z and summing them together yields the expression for L^2 in spherical polar coordinates:

$$L^2 = -\hbar^2\left[\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) + \frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\phi^2}\right]$$

2. Commutation Rules for Angular Momentum Operators

Fundamental Commutation Relations

The quantum mechanical position and momentum operators satisfy the canonical commutation relations:

$$[x_i, p_j] = i\hbar\delta_{ij}$$

$$[x_i, x_j] = 0$$

$$[p_i, p_j] = 0$$

where $x_i, x_j \in \{x, y, z\}$ and $p_i, p_j \in \{p_x, p_y, p_z\}$. We use these fundamental relations to evaluate the commutator of the Cartesian components of angular momentum.

To derive the commutator $[L_x, L_y]$, we substitute the Cartesian definitions $L_x = yp_z - zp_y$ and $L_y = zp_x - xp_z$:

$$[L_x, L_y] = [yp_z - zp_y, zp_x - xp_z]$$

Using the linearity property of commutators, $[A - B, C - D] = [A, C] - [A, D] - [B, C] + [B, D]$, we expand this into four terms:

$$[L_x, L_y] = [yp_z, zp_x] - [yp_z, xp_z] - [zp_y, zp_x] + [zp_y, xp_z]$$

We evaluate each term separately using the identity $[AB, CD] = A[B, C]D + AC[B, D] + [A, C]DB + C[A, D]B$. For the first term $[yp_z, zp_x]$, noting that y and p_x commute with z and p_z, we simplify:

$$[yp_z, zp_x] = y[p_z, z]p_x$$

Using the canonical commutation relation $[p_z, z] = -i\hbar$, we substitute this identity into the first term:

$$[yp_z, zp_x] = -i\hbar y p_x$$

For the second term $[yp_z, xp_z]$, since all operators involved (y, p_z, x) commute with each other:

$$[yp_z, xp_z] = 0$$

For the third term $[zp_y, zp_x]$, since all operators involved (z, p_y, p_x) commute with each other:

$$[zp_y, zp_x] = 0$$

For the fourth term $[zp_y, xp_z]$, noting that x and p_y commute with z and p_z, we simplify:

$$[zp_y, xp_z] = x[z, p_z]p_y$$

Using the canonical commutation relation $[z, p_z] = i\hbar$, we substitute this identity into the fourth term:

$$[zp_y, xp_z] = i\hbar x p_y$$

Substituting the evaluated results for all four terms back into the expanded commutator for $[L_x, L_y]$ gives:

$$[L_x, L_y] = -i\hbar y p_x + 0 - 0 + i\hbar x p_y$$

Factoring out $i\hbar$ from the remaining terms yields:

$$[L_x, L_y] = i\hbar(xp_y - yp_x)$$

Recognizing that $xp_y - yp_x = L_z$, we replace the quantity in parentheses with L_z:

$$[L_x, L_y] = i\hbar L_z$$

By cyclic permutation of the indices $(x \to y \to z \to x)$, the derivation for the commutator $[L_y, L_z]$ gives:

$$[L_y, L_z] = i\hbar L_x$$

Performing another cyclic permutation $(y \to z \to x \to y)$, the derivation for the commutator $[L_z, L_x]$ gives:

$$[L_z, L_x] = i\hbar L_y$$

These three commutation relations can be written concisely in compact index notation using the Levi-Civita symbol ϵ_{ijk}, where $i, j, k \in \{1, 2, 3\}$ correspond to $\{x, y, z\}$:

$$[L_i, L_j] = i\hbar \epsilon_{ijk} L_k$$

Commutation of L^2 with Components of Angular Momentum

We now examine whether the total angular momentum squared operator, $L^2 = L_x^2 + L_y^2 + L_z^2$, commutes with its individual components.

Consider the commutator $[L^2, L_z]$. Substituting $L^2 = L_x^2 + L_y^2 + L_z^2$ gives:

$$[L^2, L_z] = [L_x^2 + L_y^2 + L_z^2, L_z]$$

Using the distributive property of commutators, we write this as a sum of three terms:

$$[L^2, L_z] = [L_x^2, L_z] + [L_y^2, L_z] + [L_z^2, L_z]$$

Since any operator commutes with its own power, $[L_z^2, L_z] = 0$, reducing the expression to:

$$[L^2, L_z] = [L_x^2, L_z] + [L_y^2, L_z]$$

We evaluate the term $[L_x^2, L_z]$ using the operator identity $[A^2, B] = A[A, B] + [A, B]A$:

$$[L_x^2, L_z] = L_x[L_x, L_z] + [L_x, L_z]L_x$$

Since $[L_z, L_x] = i\hbar L_y$, the anti-symmetry property of commutators gives $[L_x, L_z] = -i\hbar L_y$.
Substituting this identity into the expression yields:

$$[L_x^2, L_z] = L_x(-i\hbar L_y) + (-i\hbar L_y)L_x$$

Factoring out $-i\hbar$ gives:

$$[L_x^2, L_z] = -i\hbar(L_x L_y + L_y L_x)$$

Next, we evaluate the term $[L_y^2, L_z]$ using the identity $[A^2, B] = A[A, B] + [A, B]A$:

$$[L_y^2, L_z] = L_y[L_y, L_z] + [L_y, L_z]L_y$$

Substituting the derived commutation relation $[L_y, L_z] = i\hbar L_x$ gives:

$$[L_y^2, L_z] = L_y(i\hbar L_x) + (i\hbar L_x)L_y$$

Factoring out $i\hbar$ gives:

$$[L_y^2, L_z] = i\hbar(L_y L_x + L_x L_y)$$

Adding $[L_x^2, L_z]$ and $[L_y^2, L_z]$ together to determine $[L^2, L_z]$ yields:

$$[L^2, L_z] = -i\hbar(L_x L_y + L_y L_x) + i\hbar(L_y L_x + L_x L_y)$$

Grouping the terms inside brackets gives:

$$[L^2, L_z] = i\hbar\left(-L_x L_y - L_y L_x + L_y L_x + L_x L_y\right)$$

Simplifying the sum inside the parentheses yields zero:

$$[L^2, L_z] = 0$$

By cyclic symmetry of the definition of $L^2 = L_x^2 + L_y^2 + L_z^2$, identical derivations apply to $[L^2, L_x]$ and $[L^2, L_y]$, yielding:

$$[L^2, L_x] = 0$$

$$[L^2, L_y] = 0$$

Thus, L^2 commutes with all three Cartesian components:

$$[L^2, L_x] = [L^2, L_y] = [L^2, L_z] = 0$$

Simultaneous Measurement of L^2 and L_z

In quantum mechanics, two physical observables can be simultaneously measured with exact precision (meaning they share a complete set of simultaneous eigenfunctions) if and only if their corresponding operators commute.

Because $[L^2, L_z] = 0$, the operators L^2 and L_z commute. Consequently, a quantum system can exist in a state that is an eigenstate of both L^2 and L_z simultaneously. Thus, the total angular momentum magnitude squared L^2 and its projection along one chosen spatial axis (by standard convention, the z-axis, L_z) can be measured simultaneously with defined values.

In contrast, because the Cartesian components do not commute with one another ($[L_x, L_y] \neq 0$, $[L_y, L_z] \neq 0$, $[L_z, L_x] \neq 0$), no two components of angular momentum can be measured simultaneously with exact precision. Therefore, a physical state can be specified by simultaneous eigenvalues of L^2 and only one component of angular momentum, usually chosen as L_z.

3. Eigenvalues and Eigenfunctions of L^2 and L_z

In quantum mechanics, the orbital angular momentum operators $\hat{L}^2$ and $\hat{L}_z$ commute with each other, $[\hat{L}^2, \hat{L}_z] = 0$. Consequently, they share a complete set of simultaneous eigenfunctions. To determine these eigenfunctions and their corresponding eigenvalues, we express the operators in spherical polar coordinates (r, θ, ϕ).

The z-component of orbital angular momentum in spherical polar coordinates is given by:

$$\hat{L}_z = -i\hbar\frac{\partial}{\partial\phi}$$

The square of the total orbital angular momentum operator $\hat{L}^2$ in spherical polar coordinates is given by:

$$\hat{L}^2 = -\hbar^2\left[\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) + \frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\phi^2}\right]$$

We represent the common eigenfunction as $Y(\theta,\phi)$ and write the eigenvalue equation for $\hat{L}_z$ as:

$$\hat{L}_z Y(\theta,\phi) = m_l\hbar Y(\theta,\phi)$$

where $m_l\hbar$ is the eigenvalue of $\hat{L}_z$. Since the operator $\hat{L}_z$ acts only on the azimuthal angle ϕ, we can separate the eigenfunction $Y(\theta,\phi)$ into a product of a polar function $\Theta(\theta)$ and an azimuthal function $\Phi(\phi)$:

$$Y(\theta,\phi) = \Theta(\theta)\Phi(\phi)$$

Substituting this product into the eigenvalue equation for $\hat{L}_z$ gives:

$$-i\hbar\frac{\partial}{\partial\phi}[\Theta(\theta)\Phi(\phi)] = m_l\hbar\Theta(\theta)\Phi(\phi)$$

Since $\Theta(\theta)$ does not depend on ϕ, we pull it outside the derivative:

$$-i\hbar\Theta(\theta)\frac{d\Phi(\phi)}{d\phi} = m_l\hbar\Theta(\theta)\Phi(\phi)$$

Dividing both sides of the equation by $-i\hbar\Theta(\theta)$ yields:

$$\frac{d\Phi(\phi)}{d\phi} = \frac{m_l\hbar}{-i\hbar}\Phi(\phi)$$

Simplifying the fraction on the right-hand side using $1/(-i) = i$ gives:

$$\frac{d\Phi(\phi)}{d\phi} = im_l\Phi(\phi)$$

Integrating both sides with respect to ϕ leads to:

$$\int\frac{1}{\Phi(\phi)}d\Phi(\phi) = im_l\int d\phi$$

$$\ln\Phi(\phi) = im_l\phi + C$$

Taking the exponential of both sides produces the solution:

$$\Phi(\phi) = Ae^{im_l\phi}$$

where $A = e^C$ is a normalization constant. Physical wavefunctions must be single-valued under a full rotation of 2π around the z-axis, which requires:

$$\Phi(\phi + 2\pi) = \Phi(\phi)$$

Substituting our exponential solution into this single-valuedness condition gives:

$$Ae^{im_l(\phi+2\pi)} = Ae^{im_l\phi}$$

Dividing both sides by $Ae^{im_l\phi}$ yields:

$$e^{i2\pi m_l} = 1$$

Using Euler's formula, this condition implies $\cos(2\pi m_l) + i\sin(2\pi m_l) = 1$, which holds if and only if m_l is an integer:

$$m_l = 0, \pm 1, \pm 2, \ldots$$

We determine the normalization constant A by normalizing $\Phi(\phi)$ over the interval $[0, 2\pi]$:

$$\int_0^{2\pi} |\Phi(\phi)|^2 d\phi = 1$$

Substituting $\Phi(\phi) = Ae^{im_l\phi}$ into the integral gives:

$$\int_0^{2\pi} |A|^2 e^{-im_l\phi} e^{im_l\phi} d\phi = 1$$

$$|A|^2 \int_0^{2\pi} d\phi = 1$$

$$|A|^2 (2\pi) = 1$$

$$|A| = \frac{1}{\sqrt{2\pi}}$$

Thus, the normalized azimuthal eigenfunction is:

$$\Phi_{m_l}(\phi) = \frac{1}{\sqrt{2\pi}} e^{im_l\phi}$$

Next, we address the eigenvalue equation for the total angular momentum operator $\hat{L}^2$:

$$\hat{L}^2 Y(\theta, \phi) = \lambda\hbar^2 Y(\theta, \phi)$$

where $\lambda\hbar^2$ denotes the eigenvalue of $\hat{L}^2$. Substituting the explicit coordinate form of $\hat{L}^2$ and $Y(\theta, \phi) = \Theta(\theta)\Phi_{m_l}(\phi)$ gives:

$$-\hbar^2\left[\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right)+\frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\phi^2}\right][\Theta(\theta)\Phi_{m_l}(\phi)]=\lambda\hbar^2\Theta(\theta)\Phi_{m_l}(\phi)$$

Dividing both sides by $-\hbar^2$ yields:

$$\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial[\Theta(\theta)\Phi_{m_l}(\phi)]}{\partial\theta}\right)+\frac{1}{\sin^2\theta}\frac{\partial^2[\Theta(\theta)\Phi_{m_l}(\phi)]}{\partial\phi^2}=-\lambda\Theta(\theta)\Phi_{m_l}(\phi)$$

Carrying out the partial derivatives:

$$\Phi_{m_l}(\phi)\frac{1}{\sin\theta}\frac{d}{d\theta}\left(\sin\theta\frac{d\Theta(\theta)}{d\theta}\right)+\Theta(\theta)\frac{1}{\sin^2\theta}\frac{d^2\Phi_{m_l}(\phi)}{d\phi^2}=-\lambda\Theta(\theta)\Phi_{m_l}(\phi)$$

Evaluating the second derivative of $\Phi_{m_l}(\phi)=\frac{1}{\sqrt{2\pi}}e^{im_l\phi}$ with respect to ϕ:

$$\frac{d^2\Phi_{m_l}(\phi)}{d\phi^2}=-m_l^2\Phi_{m_l}(\phi)$$

Substituting this second derivative back into the differential equation gives:

$$\Phi_{m_l}(\phi)\frac{1}{\sin\theta}\frac{d}{d\theta}\left(\sin\theta\frac{d\Theta(\theta)}{d\theta}\right)-\Theta(\theta)\frac{m_l^2}{\sin^2\theta}\Phi_{m_l}(\phi)=-\lambda\Theta(\theta)\Phi_{m_l}(\phi)$$

Dividing the entire equation by $\Phi_{m_l}(\phi)$ yields:

$$\frac{1}{\sin\theta}\frac{d}{d\theta}\left(\sin\theta\frac{d\Theta(\theta)}{d\theta}\right)-\frac{m_l^2}{\sin^2\theta}\Theta(\theta)=-\lambda\Theta(\theta)$$

Rearranging terms by transferring $-\lambda\Theta(\theta)$ to the left-hand side gives:

$$\frac{1}{\sin\theta}\frac{d}{d\theta}\left(\sin\theta\frac{d\Theta(\theta)}{d\theta}\right)+\left(\lambda-\frac{m_l^2}{\sin^2\theta}\right)\Theta(\theta)=0$$

To solve this equation, we introduce a variable transformation $x=\cos\theta$. Using the chain rule, the derivative with respect to θ becomes:

$$\frac{d}{d\theta}=\frac{dx}{d\theta}\frac{d}{dx}=-\sin\theta\frac{d}{dx}$$

We also express $\sin^2\theta$ in terms of x:

$$\sin^2\theta=1-\cos^2\theta=1-x^2$$

Substituting these into the term $\sin\theta\frac{d\Theta}{d\theta}$ yields:

$$\sin\theta\frac{d\Theta}{d\theta}=\sin\theta\left(-\sin\theta\frac{d\Theta}{dx}\right)=-\sin^2\theta\frac{d\Theta}{dx}=-(1-x^2)\frac{d\Theta}{dx}$$

Applying $\frac{1}{\sin\theta}\frac{d}{d\theta}$ to this expression gives:

$$\frac{1}{\sin\theta}\frac{d}{d\theta}\left(\sin\theta\frac{d\Theta}{d\theta}\right) = \frac{1}{\sin\theta}\left(-\sin\theta\frac{d}{dx}\right)\left[-(1-x^2)\frac{d\Theta}{dx}\right] = \frac{d}{dx}\left[(1-x^2)\frac{d\Theta}{dx}\right]$$

Substituting this transformed derivative back into the differential equation gives:

$$\frac{d}{dx}\left[(1-x^2)\frac{d\Theta}{dx}\right] + \left(\lambda - \frac{m_l^2}{1-x^2}\right)\Theta = 0$$

Expanding the first term using the product rule leads to:

$$(1-x^2)\frac{d^2\Theta}{dx^2} - 2x\frac{d\Theta}{dx} + \left(\lambda - \frac{m_l^2}{1-x^2}\right)\Theta = 0$$

This is the associated Legendre differential equation. For the eigenfunctions $\Theta(x)$ to be physically acceptable, they must remain finite and well-behaved at the boundaries $x = \pm 1$ (which correspond to the poles $\theta = 0$ and $\theta = \pi$). Quantum mechanical boundary conditions demand that the parameter λ must satisfy:

$$\lambda = l(l+1)$$

where the orbital quantum number l takes non-negative integer values:

$$l = 0, 1, 2, 3, \ldots$$

For a given value of l, the integer m_l is restricted to $(2l+1)$ values:

$$m_l = -l, -(l-1), \ldots, 0, \ldots, (l-1), l$$

The regular solutions to the associated Legendre equation are proportional to the associated Legendre polynomials $P_l^{m_l}(\cos\theta)$. The normalized polar eigenfunctions are:

$$\Theta_{l,m_l}(\theta) = \sqrt{\frac{2l+1}{2}\frac{(l-m_l)!}{(l+m_l)!}}P_l^{m_l}(\cos\theta)$$

Combining the normalized polar function $\Theta_{l,m_l}(\theta)$ and the normalized azimuthal function $\Phi_{m_l}(\phi)$ yields the simultaneous eigenfunctions of $\hat{L}^2$ and $\hat{L}_z$, known as the spherical harmonics $Y_l^{m_l}(\theta,\phi)$:

$$Y_l^{m_l}(\theta,\phi) = \Theta_{l,m_l}(\theta)\Phi_{m_l}(\phi)$$

Substituting the explicit formulas for $\Theta_{l,m_l}(\theta)$ and $\Phi_{m_l}(\phi)$:

$$Y_l^{m_l}(\theta,\phi) = \sqrt{\frac{2l+1}{2}\frac{(l-m_l)!}{(l+m_l)!}}P_l^{m_l}(\cos\theta)\cdot\frac{1}{\sqrt{2\pi}}e^{im_l\phi}$$

Combining the numerical constants under a single square root yields:

$$Y_l^{m_l}(\theta,\phi) = (-1)^{m_l}\sqrt{\frac{2l+1}{4\pi}\frac{(l-m_l)!}{(l+m_l)!}}P_l^{m_l}(\cos\theta)e^{im_l\phi}$$

where the factorials use $|m_l|$ when m_l is negative, and $Y_l^{-m_l} = (-1)^{m_l}(Y_l^{m_l})^*$; $(-1)^{m_l}$ is the standard Condon-Shortley phase factor. The simultaneous eigenvalue equations for the spherical harmonics are:

$$\hat{L}^2 Y_l^{m_l}(\theta,\phi) = l(l+1)\hbar^2 Y_l^{m_l}(\theta,\phi)$$

$$\hat{L}_z Y_l^{m_l}(\theta,\phi) = m_l \hbar Y_l^{m_l}(\theta,\phi)$$

The magnitude of the total orbital angular momentum vector $|\mathbf{L}|$ is determined by the eigenvalue of $\hat{L}^2$:

$$|\mathbf{L}| = \sqrt{l(l+1)}\hbar$$

The orbital quantum number $l = 0, 1, 2, \ldots$ determines the total angular momentum magnitude $\sqrt{l(l+1)}\hbar$, while $m_l = -l, \ldots, +l$ determines the z-component $L_z = m_l\hbar$.

Spatial Quantization

Spatial quantization (or space quantization) refers to the phenomenon where the orbital angular momentum vector $\mathbf{L}$ cannot point in arbitrary directions in space, but is restricted to a discrete set of orientations relative to a chosen reference axis (conventionally taken as the z-axis).

The angle θ between the angular momentum vector $\mathbf{L}$ and the z-axis is defined classically by:

$$\cos\theta = \frac{L_z}{|\mathbf{L}|}$$

Substituting the quantum mechanical eigenvalues $L_z = m_l\hbar$ and $|\mathbf{L}| = \sqrt{l(l+1)}\hbar$ into this ratio gives:

$$\cos\theta = \frac{m_l\hbar}{\sqrt{l(l+1)}\hbar}$$

Canceling $\hbar$ from the numerator and denominator yields:

$$\cos\theta = \frac{m_l}{\sqrt{l(l+1)}}$$

Since the magnetic quantum number m_l can take only $(2l+1)$ discrete integer values for a given orbital quantum number l:

$$m_l = -l, -(l-1), \ldots, 0, \ldots, (l-1), l$$

the angle θ can likewise assume only $(2l+1)$ discrete values. The angular momentum vector is therefore quantized in orientation, constrained to lie along $(2l+1)$ discrete cones relative to the z-axis. Furthermore, because $|m_l| \leq l < \sqrt{l(l+1)}$, the magnitude $|m_l/\sqrt{l(l+1)}|$ is strictly less than 1. Consequently, $\cos\theta$ can never equal $+1$ or -1, meaning the vector $\mathbf{L}$ can never align completely along the z-axis.

4. Ladder Operators and Their Commutation Relations

In quantum mechanics, the study of angular momentum is greatly simplified by defining non-Hermitian operators known as ladder operators, or raising and lowering operators. The raising operator L_+ and the lowering operator L_- are defined as linear combinations of the Cartesian components of angular momentum L_x and L_y:

$$L_+ = L_x + iL_y$$

$$L_- = L_x - iL_y$$

Taking the Hermitian adjoint of these operators reveals their adjoint relationship:

$$(L_+)^\dagger = (L_x + iL_y)^\dagger = L_x^\dagger - iL_y^\dagger = L_x - iL_y = L_-$$

$$(L_-)^\dagger = (L_x - iL_y)^\dagger = L_x^\dagger + iL_y^\dagger = L_x + iL_y = L_+$$

Commutation Relations of Ladder Operators

To determine how these operators interact with the standard angular momentum operators, we calculate their fundamental commutation relations using the standard commutation relations $[L_x, L_y] = i\hbar L_z$, $[L_y, L_z] = i\hbar L_x$, $[L_z, L_x] = i\hbar L_y$, and $[L^2, L_x] = [L^2, L_y] = [L^2, L_z] = 0$.

First, we evaluate the commutator $[L_z, L_+]$ by expanding L_+ into its components:

$$[L_z, L_+] = [L_z, L_x + iL_y]$$

Using the linearity of the commutator, this splits into two terms:

$$[L_z, L_+] = [L_z, L_x] + i[L_z, L_y]$$

Substituting the fundamental angular momentum commutation relations $[L_z, L_x] = i\hbar L_y$ and $[L_z, L_y] = -i\hbar L_x$ yields:

$$[L_z, L_+] = i\hbar L_y + i(-i\hbar L_x)$$

Simplifying the second term gives:

$$[L_z, L_+] = i\hbar L_y + \hbar L_x$$

Factoring out $\hbar$ brings this to the form:

$$[L_z, L_+] = \hbar(L_x + iL_y) = +\hbar L_+$$

Next, we evaluate the commutator $[L_z, L_-]$ by expanding L_-:

$$[L_z, L_-] = [L_z, L_x - iL_y]$$

Using linearity, we split the commutator:

$$[L_z, L_-] = [L_z, L_x] - i[L_z, L_y]$$

Substituting $[L_z, L_x] = i\hbar L_y$ and $[L_z, L_y] = -i\hbar L_x$ gives:

$$[L_z, L_-] = i\hbar L_y - i(-i\hbar L_x)$$

Simplifying the imaginary product gives:

$$[L_z, L_-] = i\hbar L_y - \hbar L_x$$

Factoring out $-\hbar$ yields:

$$[L_z, L_-] = -\hbar(L_x - iL_y) = -\hbar L_-$$

We now calculate the commutator $[L_+, L_-]$ between the raising and lowering operators themselves:

$$[L_+, L_-] = [L_x + iL_y, L_x - iL_y]$$

Expanding the commutator gives four terms:

$$[L_+, L_-] = [L_x, L_x] - i[L_x, L_y] + i[L_y, L_x] + [L_y, L_y]$$

Since any operator commutes with itself, $[L_x, L_x] = 0$ and $[L_y, L_y] = 0$. Furthermore, using $[L_y, L_x] = -[L_x, L_y]$, the expression becomes:

$$[L_+, L_-] = 0 - i[L_x, L_y] - i[L_x, L_y] + 0 = -2i[L_x, L_y]$$

Substituting $[L_x, L_y] = i\hbar L_z$ yields:

$$[L_+, L_-] = -2i(i\hbar L_z) = 2\hbar L_z$$

Finally, we consider the commutators of L^2 with L_+ and L_-. Expanding $[L^2, L_+]$ gives:

$$[L^2, L_+] = [L^2, L_x + iL_y] = [L^2, L_x] + i[L^2, L_y]$$

Since L^2 commutes with each individual Cartesian component ($[L^2, L_x] = 0$ and $[L^2, L_y] = 0$), we have:

$$[L^2, L_+] = 0 + i(0) = 0$$

Similarly, for $[L^2, L_-]$:

$$[L^2, L_-] = [L^2, L_x - iL_y] = [L^2, L_x] - i[L^2, L_y] = 0 - 0 = 0$$

Summarizing these key commutation relations:

$$[L_z, L_+] = +\hbar L_+$$

$$[L_z, L_-] = -\hbar L_-$$

$$[L_+, L_-] = 2\hbar L_z$$

$$[L^2, L_+] = [L^2, L_-] = 0$$

Action of Ladder Operators on Spherical Harmonics

Let $Y_l^{m_l}$ be a simultaneous eigenfunction of L^2 and L_z, satisfying the eigenvalue equations:

$$L^2 Y_l^{m_l} = l(l+1)\hbar^2 Y_l^{m_l}$$

$$L_z Y_l^{m_l} = m_l \hbar Y_l^{m_l}$$

To understand the action of $L_\pm$ on $Y_l^{m_l}$, we first apply L^2 to the state $L_\pm Y_l^{m_l}$. Since $[L^2, L_\pm] = 0$, the operators commute:

$$L^2(L_\pm Y_l^{m_l}) = L_\pm L^2 Y_l^{m_l}$$

Substituting the eigenvalue equation for L^2 yields:

$$L^2(L_\pm Y_l^{m_l}) = L_\pm \left(l(l+1)\hbar^2 Y_l^{m_l} \right) = l(l+1)\hbar^2 (L_\pm Y_l^{m_l})$$

This demonstrates that $L_\pm Y_l^{m_l}$ remains an eigenfunction of L^2 with the exact same eigenvalue $l(l+1)\hbar^2$, meaning the quantum number l is unchanged under the action of ladder operators.

Now, we apply L_z to the state $L_\pm Y_l^{m_l}$. Using the commutation relation $L_z L_\pm - L_\pm L_z = \pm \hbar L_\pm$, we rewrite $L_z L_\pm$ as:

$$L_z L_\pm = L_\pm L_z \pm \hbar L_\pm$$

Applying this identity to $Y_l^{m_l}$ gives:

$$L_z(L_\pm Y_l^{m_l}) = (L_\pm L_z \pm \hbar L_\pm) Y_l^{m_l}$$

Distributing the operator across the eigenstate yields:

$$L_z(L_\pm Y_l^{m_l}) = L_\pm (L_z Y_l^{m_l}) \pm \hbar L_\pm Y_l^{m_l}$$

Substituting $L_z Y_l^{m_l} = m_l \hbar Y_l^{m_l}$ gives:

$$L_z(L_\pm Y_l^{m_l}) = L_\pm (m_l \hbar Y_l^{m_l}) \pm \hbar L_\pm Y_l^{m_l}$$

Factoring out the constants and the state $L_\pm Y_l^{m_l}$ results in:

$$L_z(L_\pm Y_l^{m_l}) = (m_l \pm 1)\hbar (L_\pm Y_l^{m_l})$$

This result proves that L_+ increases the eigenvalue of L_z by $\hbar$ (raising m_l to $m_l + 1$), while L_- decreases the eigenvalue of L_z by $\hbar$ (lowering m_l to $m_l - 1$). Thus, $L_\pm Y_l^{m_l}$ is proportional to $Y_l^{m_l \pm 1}$:

$$L_+ Y_l^{m_l} = C_+ Y_l^{m_l+1}$$

$$L_- Y_l^{m_l} = C_- Y_l^{m_l-1}$$

Operator Products L_-L_+ and L_+L_-

To determine the constant factors C_+ and C_-, we express the operator products L_-L_+ and L_+L_- in terms of L^2 and L_z. Evaluating L_-L_+ gives:

$$L_-L_+ = (L_x - iL_y)(L_x + iL_y) = L_x^2 + iL_xL_y - iL_yL_x + L_y^2$$

Group the real terms and rewrite the imaginary terms as a commutator:

$$L_-L_+ = L_x^2 + L_y^2 + i[L_x, L_y]$$

Substituting $L_x^2 + L_y^2 = L^2 - L_z^2$ and $[L_x, L_y] = i\hbar L_z$ yields:

$$L_-L_+ = L^2 - L_z^2 + i(i\hbar L_z) = L^2 - L_z^2 - \hbar L_z$$

Similarly, evaluating L_+L_- gives:

$$L_+L_- = (L_x + iL_y)(L_x - iL_y) = L_x^2 - iL_xL_y + iL_yL_x + L_y^2$$

Grouping the terms yields:

$$L_+L_- = L_x^2 + L_y^2 - i[L_x, L_y]$$

Substituting $L_x^2 + L_y^2 = L^2 - L_z^2$ and $[L_x, L_y] = i\hbar L_z$ gives:

$$L_+L_- = L^2 - L_z^2 - i(i\hbar L_z) = L^2 - L_z^2 + \hbar L_z$$

Derivation of Normalization Constants

To calculate C_+, we take the norm of the state $L_+Y_l^{m_l}$:

$$\langle L_+Y_l^{m_l} | L_+Y_l^{m_l} \rangle = |C_+|^2 \langle Y_l^{m_l+1} | Y_l^{m_l+1} \rangle = |C_+|^2$$

Using the adjoint relation $(L_+)^\dagger = L_-$, the left-hand side can be rewritten as:

$$\langle L_+Y_l^{m_l} | L_+Y_l^{m_l} \rangle = \langle Y_l^{m_l} | L_-L_+Y_l^{m_l} \rangle$$

Substituting $L_-L_+ = L^2 - L_z^2 - \hbar L_z$ into this inner product:

$$\langle Y_l^{m_l} | L_-L_+Y_l^{m_l} \rangle = \langle Y_l^{m_l} | (L^2 - L_z^2 - \hbar L_z) Y_l^{m_l} \rangle$$

Operating $\left(L^2 - L_z^2 - \hbar L_z\right)$ on $Y_l^{m_l}$ yields:

$$(L^2 - L_z^2 - \hbar L_z)Y_l^{m_l} = \left(l(l+1)\hbar^2 - m_l^2\hbar^2 - m_l\hbar^2\right) Y_l^{m_l}$$

Factoring out $\hbar^2$ gives:

$$(L^2 - L_z^2 - \hbar L_z)Y_l^{m_l} = \hbar^2 \left(l(l+1) - m_l(m_l+1)\right) Y_l^{m_l}$$

Because $Y_l^{m_l}$ is normalized, taking the inner product yields:

$$|C_+|^2 = \hbar^2 \left(l(l+1) - m_l(m_l+1) \right)$$

Adopting the standard positive real phase convention for the matrix elements, we obtain:

$$C_+ = \hbar \sqrt{l(l+1) - m_l(m_l+1)}$$

Therefore, the explicit action of the raising operator on a spherical harmonic is:

$$L_+ Y_l^{m_l} = \hbar \sqrt{l(l+1) - m_l(m_l+1)} Y_l^{m_l+1}$$

To calculate C_-, we take the norm of the state $L_- Y_l^{m_l}$:

$$\langle L_- Y_l^{m_l} | L_- Y_l^{m_l} \rangle = |C_-|^2 \langle Y_l^{m_l-1} | Y_l^{m_l-1} \rangle = |C_-|^2$$

Using the adjoint relation $(L_-)^\dagger = L_+$, we rewrite the left-hand side:

$$\langle L_- Y_l^{m_l} | L_- Y_l^{m_l} \rangle = \langle Y_l^{m_l} | L_+ L_- Y_l^{m_l} \rangle$$

Substituting $L_+ L_- = L^2 - L_z^2 + \hbar L_z$ into the expression:

$$\langle Y_l^{m_l} | L_+ L_- Y_l^{m_l} \rangle = \langle Y_l^{m_l} | (L^2 - L_z^2 + \hbar L_z) Y_l^{m_l} \rangle$$

Operating $\left(L^2 - L_z^2 + \hbar L_z\right)$ on $Y_l^{m_l}$ gives:

$$(L^2 - L_z^2 + \hbar L_z) Y_l^{m_l} = \left(l(l+1)\hbar^2 - m_l^2 \hbar^2 + m_l \hbar^2 \right) Y_l^{m_l}$$

Factoring out $\hbar^2$ yields:

$$(L^2 - L_z^2 + \hbar L_z) Y_l^{m_l} = \hbar^2 \left(l(l+1) - m_l(m_l-1) \right) Y_l^{m_l}$$

Since $Y_l^{m_l}$ is normalized, we equate the norms:

$$|C_-|^2 = \hbar^2 \left(l(l+1) - m_l(m_l-1) \right)$$

Choosing the standard positive real phase convention gives:

$$C_- = \hbar \sqrt{l(l+1) - m_l(m_l-1)}$$

Therefore, the explicit action of the lowering operator on a spherical harmonic is:

$$L_- Y_l^{m_l} = \hbar \sqrt{l(l+1) - m_l(m_l-1)} Y_l^{m_l-1}$$

Termination of the Ladder and Bounds on m_l

Because any physical state in Hilbert space must have a non-negative norm, the norm of $L_+ Y_l^{m_l}$ cannot be negative:

$$\langle L_+ Y_l^{m_l} | L_+ Y_l^{m_l} \rangle = \hbar^2 \left(l(l+1) - m_l(m_l+1) \right) \geq 0$$

This requires that:

$$l(l+1) - m_l(m_l+1) \geq 0$$

If repeated applications of L_+ were to increase m_l indefinitely, $m_l(m_l+1)$ would eventually exceed $l(l+1)$, causing the norm to become negative and violating the positivity requirement of Hilbert space. To prevent this, there must exist a maximum value $m_{l,\max}$ beyond which the ladder terminates, meaning the raising operator annihilates the state:

$$L_+ Y_l^{m_{l,\max}} = 0$$

Taking the squared norm of this equation gives:

$$\hbar^2 \left(l(l+1) - m_{l,\max}(m_{l,\max}+1) \right) = 0$$

Since $\hbar \neq 0$, we must have:

$$l(l+1) - m_{l,\max}(m_{l,\max}+1) = 0$$

This yields $m_{l,\max} = l$.

Similarly, the norm of $L_- Y_l^{m_l}$ must be non-negative:

$$\langle L_- Y_l^{m_l} | L_- Y_l^{m_l} \rangle = \hbar^2 \left(l(l+1) - m_l(m_l-1) \right) \geq 0$$

This requires that:

$$l(l+1) - m_l(m_l-1) \geq 0$$

To prevent m_l from decreasing indefinitely into unphysical states with negative norms, there must exist a minimum value $m_{l,\min}$ at which the lowering operator annihilates the state:

$$L_- Y_l^{m_{l,\min}} = 0$$

Taking the squared norm yields:

$$\hbar^2 \left(l(l+1) - m_{l,\min}(m_{l,\min}-1) \right) = 0$$

Solving for $m_{l,\min}$ gives:

$$m_{l,\min} = -l$$

Thus, the sequence of states generated by the ladder operators terminates at an upper bound $m_l = +l$ and a lower bound $m_l = -l$. This establishes the strict condition governing the magnetic quantum number for a given orbital quantum number l:

$$-l \leq m_l \leq +l \quad \implies \quad |m_l| \leq l$$

5. Symmetry Transformations: Space-Time Translations and Rotations

A symmetry transformation in quantum mechanics is defined as a transformation of the coordinates or state vectors that leaves the physical properties of the system, such as probability densities and matrix elements of observables, invariant. Mathematically, a physical transformation acting on state vectors is represented by a unitary operator $\hat{U}$, which preserves the inner product between states such that $\hat{U}^\dagger \hat{U} = \hat{I}$.

Space Translations

Consider a quantum mechanical state represented by the wave function $\psi(\mathbf{r})$ in coordinate space. A spatial translation shifts the system by a displacement vector $\mathbf{a}$. Under an infinitesimal translation by $\delta\mathbf{a}$, the transformed wave function $\psi'(\mathbf{r})$ evaluated at the shifted position is given by evaluating the original wave function at the unshifted position:

$$\psi'(\mathbf{r}) = \psi(\mathbf{r} - \delta\mathbf{a})$$

Expanding the right-hand side using a three-dimensional Taylor series to first order in the infinitesimal vector $\delta\mathbf{a}$ yields:

$$\psi(\mathbf{r} - \delta\mathbf{a}) = \psi(\mathbf{r}) - \delta\mathbf{a} \cdot \nabla\psi(\mathbf{r})$$

Recall that the linear momentum operator in the position representation is expressed as:

$$\hat{\mathbf{p}} = -i\hbar\nabla$$

Rearranging this operator relation gives the gradient operator in terms of the momentum operator:

$$\nabla = \frac{i}{\hbar}\hat{\mathbf{p}}$$

Substituting this expression for the gradient operator into the Taylor expansion gives:

$$\psi'(\mathbf{r}) = \psi(\mathbf{r}) - \frac{i}{\hbar}\delta\mathbf{a} \cdot \hat{\mathbf{p}}\psi(\mathbf{r})$$

Factoring out the wave function $\psi(\mathbf{r})$ allows the infinitesimal spatial translation operator $\hat{U}(\delta\mathbf{a})$ to be identified:

$$\hat{U}(\delta\mathbf{a})\psi(\mathbf{r}) = \left(\hat{I} - \frac{i}{\hbar}\delta\mathbf{a} \cdot \hat{\mathbf{p}}\right)\psi(\mathbf{r})$$

To construct the unitary operator for a finite spatial translation by a finite vector $\mathbf{a}$, the displacement is divided into N small steps of size $\mathbf{a}/N$. Taking the limit as $N \to \infty$ produces the exponential form:

$$\hat{U}(\mathbf{a}) = \lim_{N \to \infty} \left(\hat{I} - \frac{i}{\hbar} \frac{\mathbf{a}}{N} \cdot \hat{\mathbf{p}} \right)^N$$

Evaluating this limit using the definition of the exponential function yields the finite space translation operator:

$$\hat{U}(\mathbf{a}) = \exp\left(-\frac{i}{\hbar} \hat{\mathbf{p}} \cdot \mathbf{a} \right)$$

This demonstrates that the linear momentum operator $\hat{\mathbf{p}}$ is the generator of spatial translations.

Time Translations

Consider the time evolution of a state vector $|\psi(t)\rangle$. The dynamics of the quantum state from an initial time $t = 0$ to a later time t is governed by the time-dependent Schrödinger equation:

$$i\hbar \frac{\partial}{\partial t} |\psi(t)\rangle = \hat{H} |\psi(t)\rangle$$

Assuming the Hamiltonian operator $\hat{H}$ has no explicit dependence on time, this differential equation can be formally integrated over the time interval from 0 to t:

$$|\psi(t)\rangle = \exp\left(-\frac{i}{\hbar} \hat{H} t \right) |\psi(0)\rangle$$

The time translation operator $\hat{U}(t)$, which maps the state vector at time $t = 0$ to the state vector at time t, is defined as:

$$|\psi(t)\rangle = \hat{U}(t) |\psi(0)\rangle$$

Comparing the two expressions directly defines the time translation operator:

$$\hat{U}(t) = \exp\left(-\frac{i}{\hbar} \hat{H} t \right)$$

Consequently, the Hamiltonian operator $\hat{H}$ is the generator of time translations.

Rotations

Consider a spatial rotation of a physical system about an axis specified by a unit vector $\hat{\mathbf{n}}$ through an angle θ. To derive the generator of rotations, consider first an infinitesimal rotation by an angle $\delta\theta$ about the z-axis ($\hat{\mathbf{n}} = \hat{\mathbf{z}}$). Under this transformation, the Cartesian coordinates (x, y, z) change to (x', y', z') according to:

$$x' = x \cos(\delta\theta) - y \sin(\delta\theta)$$

$$y' = x \sin(\delta\theta) + y \cos(\delta\theta)$$

$$z' = z$$

For an infinitesimal angle $\delta\theta$, the trigonometric functions may be approximated as $\cos(\delta\theta) \approx 1$ and $\sin(\delta\theta) \approx \delta\theta$:

$$x' \approx x - y\delta\theta$$

$$y' \approx y + x\delta\theta$$

$$z' = z$$

The transformed wave function $\psi'(x, y, z)$ evaluated at the rotated coordinates is equal to the original wave function evaluated at the original coordinates:

$$\psi'(x, y, z) = \psi(x + y\delta\theta, y - x\delta\theta, z)$$

Expanding this function to first order in $\delta\theta$ using a Taylor series gives:

$$\psi'(x, y, z) = \psi(x, y, z) + \delta\theta \left(y\frac{\partial}{\partial x} - x\frac{\partial}{\partial y} \right) \psi(x, y, z)$$

Recall the definition of the z-component of the orbital angular momentum operator:

$$\hat{L}_z = -i\hbar \left(x\frac{\partial}{\partial y} - y\frac{\partial}{\partial x} \right)$$

Dividing this relation by $-i\hbar$ gives the differential operator combination:

$$y\frac{\partial}{\partial x} - x\frac{\partial}{\partial y} = -\frac{i}{\hbar}\hat{L}_z$$

Substituting this identity into the Taylor expansion yields:

$$\psi'(x, y, z) = \psi(x, y, z) - \frac{i}{\hbar}\delta\theta\hat{L}_z\psi(x, y, z)$$

Factoring out the original wave function gives the infinitesimal rotation operator about the z-axis:

$$\hat{U}_{\hat{\mathbf{z}}}(\delta\theta) = \hat{I} - \frac{i}{\hbar}\delta\theta\hat{L}_z$$

Generalizing this result to an arbitrary axis defined by the unit vector $\hat{\mathbf{n}}$ gives the infinitesimal rotation operator:

$$\hat{U}_{\hat{\mathbf{n}}}(\delta\theta) = \hat{I} - \frac{i}{\hbar}\delta\theta \left(\hat{\mathbf{n}} \cdot \hat{\mathbf{L}} \right)$$

A finite rotation by an angle θ about the axis $\hat{\mathbf{n}}$ is obtained by applying N successive infinitesimal rotations of magnitude θ/N and taking the limit $N \to \infty$:

$$\hat{U}_{\hat{\mathbf{n}}}(\theta) = \lim_{N \to \infty} \left(\hat{I} - \frac{i}{\hbar}\frac{\theta}{N}\hat{\mathbf{n}} \cdot \hat{\mathbf{L}} \right)^N$$

Evaluating this limit gives the finite rotation operator:

$$\hat{U}_{\hat{\mathbf{n}}}(\theta) = \exp\left(-\frac{i}{\hbar}\theta\hat{\mathbf{n}}\cdot\hat{\mathbf{L}}\right)$$

This proves that the angular momentum operator $\hat{\mathbf{L}}$ is the generator of rotations.

6. Invariance Under Transformations and Conservation Laws

In quantum mechanics, a symmetry transformation leaves the physics of a system invariant if the transformation leaves the Hamiltonian of the system unchanged. A continuous unitary transformation operator $\hat{U}$ associated with a generator $\hat{G}$ and continuous real parameter η is written as:

$$\hat{U} = \exp\left(-\frac{i}{\hbar}\eta\hat{G}\right)$$

The Hamiltonian $\hat{H}$ is invariant under the unitary transformation $\hat{U}$ if transforming the Hamiltonian leaves it unchanged:

$$\hat{U}^\dagger\hat{H}\hat{U} = \hat{H}$$

Multiplying both sides from the left by $\hat{U}$ yields:

$$\hat{H}\hat{U} = \hat{U}\hat{H}$$

Expressing this relation as a commutator gives the condition for invariance:

$$[\hat{H}, \hat{U}] = 0$$

For an infinitesimal transformation parameter $\delta\eta$, the unitary operator is expanded to first order as $\hat{U} \approx \hat{I} - \frac{i}{\hbar}\delta\eta\hat{G}$. Substituting this into the commutator condition gives:

$$\left[\hat{H}, \hat{I} - \frac{i}{\hbar}\delta\eta\hat{G}\right] = 0$$

Since the identity operator commutes with any operator, expanding the commutator reduces this directly to:

$$-\frac{i}{\hbar}\delta\eta[\hat{H}, \hat{G}] = 0$$

Thus, the condition for the system's Hamiltonian to be invariant under the symmetry transformation generated by $\hat{G}$ is that the generator commutes with the Hamiltonian:

$$[\hat{H}, \hat{G}] = 0$$

Time Dependency of Expectation Values and Conservation Condition

To establish the quantum form of Noether's theorem, consider the time rate of change of the expectation value of an arbitrary Hermitian operator $\hat{G}$ in a state $|\psi(t)\rangle$:

$$\frac{d}{dt}\langle \hat{G} \rangle = \frac{d}{dt}\langle \psi(t)|\hat{G}|\psi(t)\rangle$$

Applying the product rule of differentiation to the inner product yields:

$$\frac{d}{dt}\langle \hat{G} \rangle = \left(\frac{\partial}{\partial t}\langle \psi(t)|\right)\hat{G}|\psi(t)\rangle + \left\langle \psi(t)\left|\frac{\partial \hat{G}}{\partial t}\right|\psi(t)\right\rangle + \langle \psi(t)|\hat{G}\left(\frac{\partial}{\partial t}|\psi(t)\rangle\right)$$

From the time-dependent Schrödinger equation, the derivative of the state vector with respect to time is:

$$\frac{\partial}{\partial t}|\psi(t)\rangle = -\frac{i}{\hbar}\hat{H}|\psi(t)\rangle$$

Taking the Hermitian conjugate of the Schrödinger equation gives the time derivative of the dual vector:

$$\frac{\partial}{\partial t}\langle \psi(t)| = \frac{i}{\hbar}\langle \psi(t)|\hat{H}$$

Substituting these time derivatives back into the expression for $\frac{d}{dt}\langle \hat{G} \rangle$ gives:

$$\frac{d}{dt}\langle \hat{G} \rangle = \frac{i}{\hbar}\langle \psi(t)|\hat{H}\hat{G}|\psi(t)\rangle + \left\langle \frac{\partial \hat{G}}{\partial t}\right\rangle - \frac{i}{\hbar}\langle \psi(t)|\hat{G}\hat{H}|\psi(t)\rangle$$

Combining the first and third terms on the right-hand side produces the expectation value of the commutator $[\hat{H}, \hat{G}]$:

$$\frac{d}{dt}\langle \hat{G} \rangle = \frac{i}{\hbar}\langle [\hat{H}, \hat{G}] \rangle + \left\langle \frac{\partial \hat{G}}{\partial t}\right\rangle$$

If the observable operator $\hat{G}$ has no explicit dependence on time, the partial derivative term vanishes:

$$\frac{\partial \hat{G}}{\partial t} = 0$$

Under this condition, the rate of change of the expectation value simplifies to:

$$\frac{d}{dt}\langle \hat{G} \rangle = \frac{i}{\hbar}\langle [\hat{H}, \hat{G}] \rangle$$

If the Hamiltonian is invariant under the transformation generated by $\hat{G}$, then $[\hat{H}, \hat{G}] = 0$. Substituting this zero commutator into the rate of change equation gives:

$$\frac{d}{dt}\langle \hat{G} \rangle = 0$$

This result demonstrates that when the Hamiltonian commutes with a time-independent generator $\hat{G}$, the expectation value of $\hat{G}$ is constant in time, meaning that $\hat{G}$ is a conserved quantity or a constant of motion.

Quantum Noether's Theorem and Specific Conservation Laws

Noether's theorem in quantum form establishes that every continuous symmetry of the Hamiltonian corresponds to a conserved observable, where the conserved observable is the generator of that symmetry transformation.

1. Invariance under Space Translation and Linear Momentum Conservation

If a physical system is invariant under spatial translations by any vector $\mathbf{a}$, the time-independent Hamiltonian must commute with the spatial translation operator $\hat{U}(\mathbf{a})$:

$$[\hat{H}, \hat{U}(\mathbf{a})] = 0$$

Since $\hat{\mathbf{p}}$ is the generator of spatial translations, this commutativity condition directly implies that the Hamiltonian commutes with each component of the linear momentum operator:

$$[\hat{H}, \hat{\mathbf{p}}] = 0$$

Applying the conservation equation $\frac{d}{dt}\langle \hat{\mathbf{p}} \rangle = \frac{i}{\hbar}\langle [\hat{H}, \hat{\mathbf{p}}] \rangle$ yields:

$$\frac{d}{dt}\langle \hat{\mathbf{p}} \rangle = 0$$

Therefore, spatial translation invariance of a system implies the conservation of linear momentum.

2. Invariance under Time Translation and Energy Conservation

If a Hamiltonian $\hat{H}$ has no explicit dependence on time, it commutes with the time translation operator $\hat{U}(t) = \exp\left(-\frac{i}{\hbar}\hat{H}t\right)$:

$$[\hat{H}, \hat{U}(t)] = 0$$

Because the Hamiltonian is its own generator for time translations, the commutator of $\hat{H}$ with itself identically vanishes:

$$[\hat{H}, \hat{H}] = 0$$

Applying the general rate of change equation to the Hamiltonian operator itself gives:

$$\frac{d}{dt}\langle \hat{H} \rangle = \frac{i}{\hbar}\langle [\hat{H}, \hat{H}] \rangle + \left\langle \frac{\partial \hat{H}}{\partial t} \right\rangle = 0$$

Therefore, time translation invariance of a system implies the conservation of total energy.

3. Invariance under Rotation and Angular Momentum Conservation

If a system is symmetric under spatial rotations about an axis $\hat{\mathbf{n}}$, its Hamiltonian commutes with the rotation operator $\hat{U}_{\hat{\mathbf{n}}}(\theta)$:

$$[\hat{H}, \hat{U}_{\hat{\mathbf{n}}}(\theta)] = 0$$

Because the angular momentum component along the axis of rotation, $\hat{L}_n = \hat{\mathbf{n}} \cdot \hat{\mathbf{L}}$, is the generator of these rotations, this invariance requires that the Hamiltonian commute with $\hat{L}_n$:

$$[\hat{H}, \hat{\mathbf{n}} \cdot \hat{\mathbf{L}}] = 0$$

Applying the equation for the time rate of change of expectation values yields:

$$\frac{d}{dt}\langle \hat{\mathbf{n}} \cdot \hat{\mathbf{L}} \rangle = \frac{i}{\hbar}\langle [\hat{H}, \hat{\mathbf{n}} \cdot \hat{\mathbf{L}}] \rangle = 0$$

If the system is isotropic (invariant under rotations about any arbitrary spatial axis), the Hamiltonian commutes with all three components of the angular momentum operator:

$$[\hat{H}, \hat{\mathbf{L}}] = 0$$

Consequently, the rate of change of the total angular momentum expectation value vanishes:

$$\frac{d}{dt}\langle \hat{\mathbf{L}} \rangle = 0$$

Therefore, rotational invariance of a physical system implies the conservation of angular momentum.

Questions

Short form first (2–3 marks), then long form (5–10 marks). **No Assam University questions were found for this unit in the paper bank** — angular momentum, ladder operators, symmetry and conservation laws are entirely untested in the bank, so this unit is a gap. The questions below are from other sources, marked as such.

Short-form questions

S1. [non-AU] Write about spatial quantization. — Univ. of Calicut, CBCSS B.Sc 5th Sem, PHY5 B07 Quantum Mechanics, Nov 2021, Q10 (2 marks).

Answer in brief: Section 3 — the angular momentum vector can only point in $(2l+1)$ discrete directions, $\cos\theta = m_l/\sqrt{l(l+1)}$, with $m_l = -l, \ldots, +l$.

S2. [non-AU] What is angular momentum quantum number? Write its possible values. — Univ. of Calicut, CBCSS B.Sc 5th Sem, PHY5 B07 Quantum Mechanics, Nov 2021, Q11.

Answer in brief: Section 3 — the orbital quantum number $l = 0, 1, 2, \ldots$ fixes the total angular momentum $\sqrt{l(l+1)}\hbar$; for each l, $m_l = -l, \ldots, +l$.

S3. [non-AU, adjacent application] A sample is placed in a 0.500 T magnetic field ... how far apart are the Zeeman components of the 480 nm spectral line? — Univ. of Calicut, CBCSS B.Sc 5th Sem, PHY5 B07 Quantum Mechanics, Nov 2021, Q18 (5 marks). Note: the Zeeman effect itself is not in this unit's syllabus; it is an application of the L_z eigenvalues of Section 3.

Answer in brief: Section 3 — adjacent m_l levels split by $\Delta E = \mu_B B$; for $B = 0.500$ T, $\Delta E \approx 4.6 \times 10^{-24}$ J, giving a wavelength separation of about 5.4×10^{-3} nm.

S4. [standard] State the commutation relations of the angular momentum operators.

Answer in brief: Section 2 — $[L_x, L_y] = i\hbar L_z$, $[L_y, L_z] = i\hbar L_x$, $[L_z, L_x] = i\hbar L_y$, and $[L^2, L_i] = 0$.

Long-form questions

L1. [standard] Derive the commutation relations for angular momentum operators, find the eigenvalues and eigenfunctions of L^2 and L_z, and explain ladder operators and their commutation relations.

What to write: Sections 2–4 in full — $[L_i, L_j] = i\hbar\epsilon_{ijk}L_k$; $L^2 Y_l^{m_l} = l(l+1)\hbar^2 Y_l^{m_l}$ and $L_z Y_l^{m_l} = m_l \hbar Y_l^{m_l}$; $L_\pm = L_x \pm iL_y$ with $[L_z, L_\pm] = \pm\hbar L_\pm$ and $[L_+, L_-] = 2\hbar L_z$.

L2. [standard] State how space-time translation and rotation invariance give conservation laws.

What to write: Sections 5–6 in full — translation generated by $\hat{p}$, time translation by $\hat{H}$, rotation by $\hat{L}$; $[\hat{H}, \hat{G}] = 0$ makes $\hat{G}$ conserved, giving conservation of momentum, energy and angular momentum.

L3. [standard] Derive the eigenvalues of total angular momentum L^2 and L_z using ladder operators, and explain spatial quantization.

What to write: Sections 3–4 in full — the ladder raises and lowers m_l by one, terminates at $m_l = \pm l$, giving $|m_l| \leq l$, total angular momentum $\sqrt{l(l+1)}\hbar$ and $(2l+1)$ orientations.

Sources

These notes were written from the unit syllabus only, as the user asked. No external PDF, textbook, or lecture notes were used. The questions are from the Assam University paper bank and other universities' papers (see the Questions section above), not from any source used for the notes.